

202. A processor has 300 distinct instructions and 70 general-purpose registers. A 32-bit instruction word has an opcode, two register operands, and an immediate operand. The number of bits available for the immediate operand field is_____

The given data,

Distinct instructions= 300

General-purpose registers = 70

opcode instruction word = 32-bit

Two register operands and an immediate operand.

Explanation:

Each instruction has 32 bits. To support 300 instructions, the opcode must contain 9-bits.

Register operand1 requires 7 bits, since the total registers are 70, Register operand 2 also requires 7 bits.

So, $32 - (9+7+7) = 9$ bits are left over for immediate operand.

Hence the correct answer is 9.

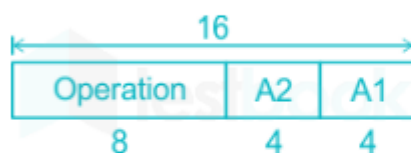
203. Only instructions with zero, one, and two addresses are supported by some CPUs. The size of an op-code is 16 bits, whereas the size of an address is 4 bits.What is the Maximum number of two address instructions?

Explanation:

The given data,

The CPU supports op-code size = 16 bits

Address size = 4 bits



We have two operands so it requires the 2×4 bits =8

And remaining $16-8$ bits can be used for two address instructions.

i.e 8 bits.

Maximum number of instructions = $2^8 = 256$

Hence the correct answer is 256.

204. A computer has an instruction size of 16 bits and has 16 programmer visible registers. Each instruction has two source and one destination operands and uses only register direct addressing. The maximum number of opcodes that this processor can have is

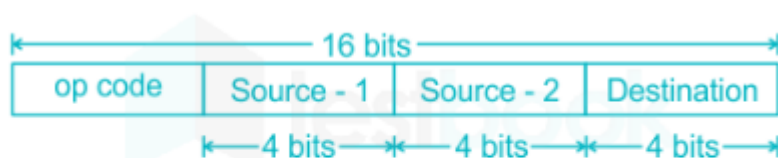
Given:

Number of registers = 16

Instruction size = 16 bits

Number of bits required to represent each register

$$n = \log_2 (16) = \log_2 2^4 = 4 \text{ bits.}$$



$$\text{No. of bits for opcode field} = 16 - 12 = 4 \text{ bits}$$

$$\therefore \text{The total number opcodes} = 2^4 = 16$$

207. A computer uses a memory unit of 512 K words of 32 bits each. A binary instruction code is stored in one word of the memory. The instruction has four parts: an addressing mode field to specify one of the two-addressing mode (direct and indirect), an operation code, a register code part to specify one of the 256 registers and an address part. How many bits are there in addressing mode part, opcode part, register code part and the address part?

Memory unit = 512 K words

1 word = 32 bits = 4 B

Binary instruction code stored in 1 word of memory

Instruction divided as follows,

Addressing mode	Operation code	Register code	Address part
-----------------	----------------	---------------	--------------

Calculation:

Addressing mode = 1 bit for direct or indirect

Register code = $\ln(256) = 8$ bits

Since, registers are already addressed, only memory needs to be addressed

Address part = $\ln(512 \text{ K}) = 19$ bits

Operation mode = $32 - 1 - 8 - 19 = 4$ bits

Therefore, bits for addressing mode part, opcode part, register code part and the address part = (1, 4, 8, 19)

210. A processor has 40 distinct instructions and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is ____.

40 distinct instructions $\rightarrow \log_2 40 = 6$ bits are needed for opcode.

24 general purpose registers $\rightarrow \log_2 24 = 5$ bits for register operand.

As, two register operands are required so, total 10 bits are needed for register operand field.

Suppose for immediate operand we require x bits.

Opcode	Register operand	Immediate operand
6 bits	10 bits	x bits

Opcode + Register operand + Immediate operand = 32

$$6 + 10 + x = 32$$

$$\therefore x = 16$$

Number of bits available for the immediate operand field (x) is 16 bits.

211. A Computer uses a memory unit with 256K word of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code and a register code part to specify one of 64 registers and an address part. How many bits are there in operation code, the register code part and the address part?

Data:

1 word = 32 bits = 4 bytes

Memory size = 256K word = 2^{18} word

Indirect bits = 1

Number of registers = $n = 64$

Formula:

Number of bits needed to present a register = $\lceil \log_2 n \rceil$

Calculation:

Number of bits needed to present a register = $\lceil \log_2 64 \rceil = 6$

Binary Instruction (32 bit):

Indirect bit (1-bit)	Operation Code (x-bits)	Register Code (6-bits)	Address Part (18-Bits)
-------------------------	----------------------------	---------------------------	---------------------------

$$1 + x + 6 + 18 = 32$$

$$\therefore x = 7$$

Therefore 7,6 and 18 bits are there in operation code, the register code part and the address part respectively.

221. A computer has 32 bit instructions and 12 bit addresses. if there are 250 two address instructions, how many one address instructions can be formulated?

Since instruction consists of opcode and address.

12 bit two address instruction will reserved 24 bit of instruction and remaining 8 bits will be used for opcode.

Total number of opcode that can be generated from 8 bit is $2^8 = 256$.

Since 250 two address instruction has already used 250 possible opcode pattern then only 6 possible pattern is left to assign to one address instruction i.e $256 - 250$.

$$6 (\text{possible opcode}) * 2^{12} (\text{bits pattern possible due to one address instruction}) = 24576.$$

Total one address instruction possible is 24576.